

1. Consider $A = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and the projective toric variety $X_A = \mathbb{P}^2$.

- (a) Check that the moment map with all weights 1 is a homeomorphism between $(X_A)_{\geq 0} = \mathbb{P}_{\geq 0}^2$ and the triangle $\text{Conv}(A) \subset \mathbb{R}^2$.
- (b) Experiment with different choices of weights and plot the images of the subsets $1 \times [0, \alpha]^2 \subset \mathbb{P}^2$ under the moment map, for various choices of $\alpha > 0$.

2. Consider the toric surface Y_A for $A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}$

- (a) Find the polynomial equation cutting out Y_A inside $\mathbb{C}^3$.
- (b) Plot Y_A and $(Y_A)_{\geq 0}$.
- (c) Using the moment map, construct a diffeomorphic parametrization $\mathbb{R}^2 \rightarrow \text{int}(\text{Cone}(A))$.

3. Consider the following polynomial in d variables: $f = \sum_{i=1}^n t^{a_i}$, where the exponents $a_i = (a_i^1, \dots, a_i^d) \in \mathbb{Z}_{\geq 0}^d$ are pairwise distinct and as usual t^{a_i} denotes $t_1^{a_i^1} \cdots t_d^{a_i^d}$. The *Newton polytope* of f , denoted $\text{Newt}(f)$, is defined to be the convex hull of the monomial support of f – that is, $\text{Newt}(f) = \text{Conv}\{a_1, \dots, a_n\} \subset \mathbb{R}^d$. Let $\text{Cone}(f) := \mathbb{R}_{\geq 0} \cdot \text{Newt}(f)$ be the cone over the Newton polytope. Prove that the mapping $\mathbb{R}^d \rightarrow \mathbb{R}^d$ given by

$$t \mapsto \left(t_1 \cdot \frac{\partial f}{\partial t_1}, \dots, t_n \cdot \frac{\partial f}{\partial t_n} \right)$$

restricts to a diffeomorphism between $\mathbb{R}_{>0}^d$ and the interior of $\text{Cone}(f)$.

Similarly, prove that the mapping $\mathbb{R}^d \rightarrow \mathbb{R}^d$ given by

$$t \mapsto \frac{1}{f} \cdot \left(t_1 \cdot \frac{\partial f}{\partial t_1}, \dots, t_n \cdot \frac{\partial f}{\partial t_n} \right)$$

restricts to a diffeomorphism between $\mathbb{R}_{>0}^d$ and the interior of $\text{Newt}(f)$.

4. Consider the matrix $A = \begin{pmatrix} 2 & 1 & 1 & 3 \\ 1 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \end{pmatrix}$.

- (a) Write down the map $\mu_A \circ \phi_A : (\mathbb{C}^*)^3 \rightarrow \mathbb{R}^3$, where μ_A denotes the moment map with all weights 1.
- (b) For each of the points $p = (7, 6, 8)$ and $(36, 26, 40)$ from the interior of $\text{Cone}(A)$, compute the preimage of p under the map from part (a) and observe that exactly one point of the preimage is in $\mathbb{R}_{>0}^3$.

5. Let X_A be a projective toric variety and let $\bar{\mu}_A : X_A \rightarrow \mathbb{R}^d$ be the algebraic moment map with all weights 1. Prove that the preimages of the vertices of $\text{Conv}(A)$ under $\bar{\mu}_A$ are the torus fixed points of X_A .
6. Prove the following two assertions from lecture.
 - (a) Recall that the positive part of an affine toric variety Y_A is given by $(Y_A)_{>0} = Y_A \cap \mathbb{R}_{>0}^n$. Show that $(Y_A)_{>0} = \phi_A(\mathbb{R}_{>0}^d)$.
 - (b) Recall that the nonnegative part of an affine toric variety Y_A is given by $(Y_A)_{\geq 0} = Y_A \cap \mathbb{R}_{\geq 0}^n$. Show that $(Y_A)_{\geq 0} = \overline{(Y_A)_{>0}}$, where the closure is taken in the Euclidean metric on $\mathbb{R}^n$.